Service Facts

YC✱120-SF-4B

Customer Property: Contains wiring and service information. Please retain.

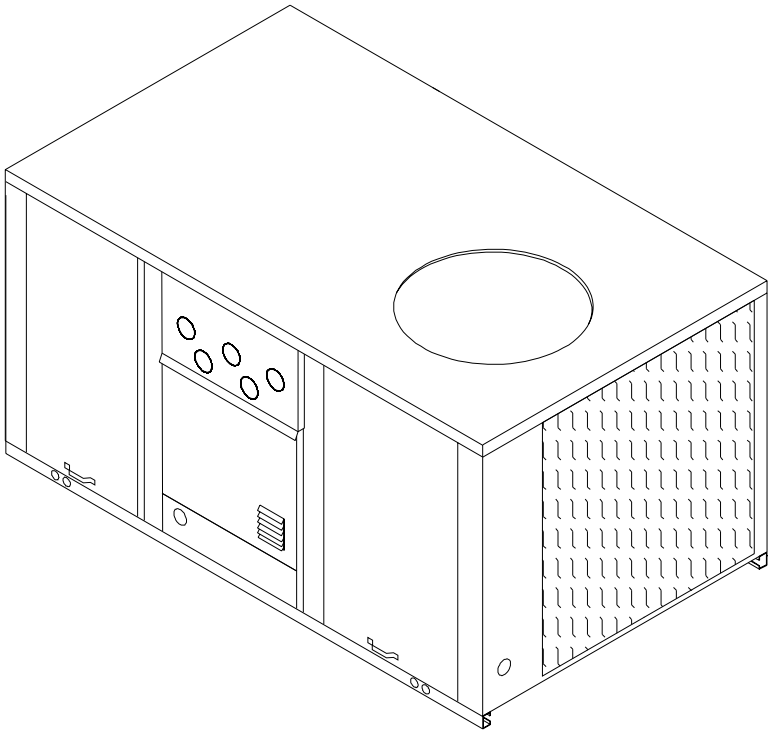
Library	Service Literature
Product Section	Unitary
Product	Package Air Conditioner
Model	YC✶
Literature Type	Service Facts
Sequence	4B
Date	June 1999
File No.	SV-UN-YC✶120-SF-4B 6/99
Supersedes	YC✶120-SF-4B 10/98

Models :

YCD120C4L0AC
YCD120C4LAAC
YCD120C4LBAC
YCD120C4LCAC
YCD120C4LFAC
YCD120C4LGAC
YCD120CWL0AC
YCD120CWLAAC
YCD120CWLBAC
YCD120CWLCAC
YCD120CWLFAC
YCD120CWLGAC
YCD120C4M0AC
YCD120C4MAAC
YCD120C4MBAC
YCD120C4MCAC
YCD120C4MFAC
YCD120C4MGAC
YCD120CWM0AC
YCD120CWMAAC
YCD120CWMBAC
YCD120CWMCAC
YCD120CWMFAC
YCD120CWMGAC
YCH120C4L0AC
YCH120C4LBAC
YCH120C4LFAC
YCH120CWL0AC
YCH120CWLBAC
YCH120CWLFAC
YCH120C4M0AC
YCH120C4MBAC
YCH120C4MFAC
YCH120CWM0AC
YCH120CWMBAC
YCH120CWMFAC

Packaged Gas/Electric

10 Ton Rooftop Units with Micro-Electronic Controls



NOTICE

Warnings and Cautions appear at appropriate locations throughout this manual.
Read these carefully

WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices and where property-damage-only accidents could occur.

Since the manufacturer has a policy of continuous product improvement,
it reserves the right to change design and specifications without notice.

Product Specifications

VOLTS/ PHASE/ HZ	460/ 3/ 60	575/ 3/ 60
A.R.I. Ratings		
Cooling Net Cap. BTUH (1)	117,000	
Part Load	70,500	
IPLV (6)	8.9	
Indoor Air Flow (CFM)	4000	
System Power (KW)	13	
EER (BTU per Watt) (2)	9	
Heating Input/Output BTUH (3)		
Low Heat	135,000/ 109,000	
Medium Heat (1st Stage)	150,000/ 122,000	
Medium Heat (2nd Stage)	205,000/ 166,000	
Temp. Rise - Min. F (L/ M)	15/ 25	
Temp. Rise - Max. F (L/ M)	45/ 55	
Gas Type **(4)**	Natural	
GAS PIPE SIZE (In.)	1/2" NPT	
POWER CONNS		
Min. Circuit Ampacity		
Standard/ Oversize Motor	31/ 33	22/ 24
Fuse Size - Max. (Amps)		
Standard/ Oversize Motor	40	25/ 30
COMPRESSOR - No Used	Two	
Volts/ Ph/ HZ	460/ 3/ 60	575/ 3/ 60
L.R. Amps (Circuit 1/ 2)	71	43
R.L. Amps (Circuit 1/ 2)	10	6.9
OUTDOOR COIL — TYPE	Hi-Performance	
Rows / F.P.I.	2/ 16	
Face Area (Sq. Ft.)	18.67	
Tube Size (In.)	0.375	
INDOOR COIL–Type	Hi-Performance	
Rows / F.P.I.	3/ 15	
Face Area (Sq. Ft.)	10.5	
Tube Size (In.)	0.375	
Refrigerant Control Device	Short Orifice	
Drain Conn. Size (In.)	3/4" PVC	
REFRIGERANT		
Charge (Lbs. of R-22) (5)	8.5 per circuit	
WEIGHT	(Approx. Lbs)	
Shipping/ Net	1191/ 984	

VOLTS/ PHASE/ HZ	460/ 3/ 60	575/ 3/ 60
INDOOR FAN		
Type/ No. Used	FC Centrifugal/ 1	
Dia. x Width (Inches)	15" x 11"	
Drive/ No. Speeds	Belt/ Variable	
INDOOR FAN MOTOR		
Motor HP/ RPM		
Standard Motor	3.0/ 1740	
Oversize Motor	5.0/ 3450	
Volts/ Ph/ Hz	460/ 3/ 60	575/ 3/ 60
L.R. Amps/ F.L. Amps		
Standard Motor	40.5/ 4.8	31/ 3.9
Oversize Motor	54.9/ 7.6	41.6/ 6.1
OUTDOOR FAN		
Type/ CFM	Propeller/ 7700	
No. Used/ Diameter	1/ 26	
Drive/ No. Speeds	Direct/ 1	
No. Motors/ HP/ RPM	1/ 1.0/ 1125	
Volts/ Ph/ Hz	460/ 1/ 60	575/ 3/ 60
L.R. Amps/ F.L. Amps	9.3/ 2.9	5.7/ 2
COMBUSTION FAN		
Type/ No. Used	Centrifugal/ 1	
Type Drive	Direct	
COMB FAN MOTOR		
Volts/ Ph/ Hz	208-230/ 1/ 60	
Motor HP (L/ M)	1/20 (.05)	
No. Speeds/ RPM		
Low Heat	1/ 3350	
Medium Heat	2/ 2800-3500	
L.R. Amps (L/ M)	1.0/ 1.4	
FILTER	Furnished	
Type	Throwaway	
Filters Size (Downflow)	16 x 20 x 2 (2)	
	20 x 20 x 2 (2)	
(Horizontal)	16 x 25 x 2 (2)	
	20 x 25 x 2 (2)	
DIMENSIONS	H x W x L	
Uncrated (In.)	45-3/8 x 50 x 88-3/8	

Footnotes:

1. Cooling Performance is rated at 95 F ambient, 80 F entering dry bulb, 67 F entering wet bulb and nominal cfm listed. ARI capacity is net and includes the effect of fan motor heat.
   Base models are certified in accordance with the Unitary Large Equipment certification program, which is based on ARI Standard 340/360-93.
   All models tested in accordance with ARI Standard 340/360-93.
2. Rated at ARI conditions and in accordance with DOE test procedures.
3. Heating performance limit settings and rating data were established and approved under laboratory test conditions using American National Standards Institute standards. Ratings shown are for elevations up to 2000 feet.
4. Convertible to LP gas with orifice change.
5. Refrigerant charge shown is a nominal value; for a more precise value see the unit nameplate.
6. Intergrated Part Load Values is based on ARI Standard 360. Ratings are 80 F ambient, 80 F entering dry bulb, and 67 F entering wet bulb at ARI rated CFM.

Performance Specifications

Evaporator Fan Performance

Downflow

	External Static Pressure (Inches of Water)																			
	0.10		0.20		0.30		0.40		0.50		0.60		0.70		0.80		0.90		1.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
	Standard Motor & Field Supplied Low Static Drive (1)																			
3200			776	1.29	807	1.39	837	1.49	866	1.60	893	1.71	920	1.82	945	1.94	970	2.05	994	2.17
3600	831	1.66	859	1.77	886	1.88	913	2.00	940	2.12	966	2.24	991	2.36	1015	2.48	1038	2.61	1061	2.74
4000	918	2.24	943	2.37	967	2.49	992	2.62	1016	2.75	1040	2.88	1064	3.01	1086	3.15	1109	3.28	1130	3.42
4400	1005	2.95	1028	3.09	1050	3.23	1073	3.37	1095	3.51	1117	3.65	1139	3.80	1160	3.94	1181	4.09	1202	4.24
	3 HP Motor & Standard Drive								(2)											
4800	1092	3.80	1113	3.96	1134	4.11	1154	4.26	1175	4.41	1195	4.57	1216	4.72	1236	4.88	1255	5.03	1275	5.19
	(2)				5 HP Oversize Motor & Drive															

	External Static Pressure (Inches of Water)																			
	1.10		1.20		1.30		1.40		1.50		1.60		1.70		1.80		1.90		2.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
	3 HP Motor & Standard Drive																			
3200	1018	2.29	1041	2.41	1063	2.53	1085	2.65	1107	2.77	1128	2.89	1149	3.02	1169	3.15	1189	3.28	1208	3.41
3600	1083	2.87	1105	3.00	1126	3.14	1147	3.27	1167	3.40	1187	3.54	1207	3.67	1226	3.81	1245	3.95	1264	4.09
4000	1151	3.56	1172	3.71	1192	3.85	1212	4.00	1231	4.15	1250	4.30	1269	4.45	1287	4.59	1306	4.74	1324	4.89
4400	1222	4.39	1241	4.54	1260	4.70	1279	4.85	1298	5.01	1316	5.17	1334	5.33	1351	5.50	1369	5.66		
4800	1294	5.35	1313	5.52	1331	5.68	5 HP Oversize Motor & Drive													

Horizontal

	External Static Pressure (Inches of Water)																			
	0.10		0.20		0.30		0.40		0.50		0.60		0.70		0.80		0.90		1.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
	Standard Motor & Field Supplied Low Static Drive (1)																			
3200			759	1.21	790	1.31	818	1.40	844	1.50	870	1.59	894	1.69	918	1.79	941	1.89	964	1.98
3600	807	1.53	838	1.65	867	1.77	893	1.88	918	1.99	942	2.10	965	2.21	987	2.32	1009	2.43	1030	2.54
4000	891	2.06	918	2.20	946	2.34	971	2.47	994	2.59	1017	2.71	1038	2.83	1059	2.95	1080	3.07	1100	3.19
4400	975	2.72	999	2.86	1025	3.02	1049	3.17	1071	3.31	1092	3.44	1113	3.57	1133	3.70	1152	3.83	1171	3.97
4800	1059	3.51	1082	3.65	1105	3.82	1128	3.99	1149	4.16	1170	4.30	1189	4.45	1208	4.59	1226	4.73	1244	4.87
	5 HP Oversize Motor & Drive																			

	External Static Pressure (Inches of Water)																			
	1.10		1.20		1.30		1.40		1.50		1.60		1.70		1.80		1.90		2.00	
CFM	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
	3 HP Standard Motor and Drive																			
3200	986	2.08	1007	2.17	1029	2.27	1049	2.37	1070	2.48	1090	2.59	1110	2.70	1130	2.81	1149	2.93	1170	3.06
3600	1051	2.65	1071	2.75	1091	2.86	1111	2.96	1130	3.07	1149	3.18	1167	3.29	1186	3.41	1204	3.53	1222	3.65
4000	1119	3.31	1138	3.44	1157	3.56	1175	3.68	1194	3.80	1211	3.92	1229	4.03	1246	4.15	1264	4.27	1281	4.39
4400	1190	4.10	1208	4.23	1226	4.37	1243	4.51	1260	4.64	1277	4.77	1294	4.91	1311	5.04	1327	5.17	1343	5.29
4800	1262	5.02	1279	5.16	1296	5.31	1313	5.46	1329	5.60								(3)		
	5 HP Oversize Motor & Drive																			

Notes:

Fan motor heat (MBH) = 3.15 x Fan BHP.

Data includes pressure Drop Due to Wet Coils and Filters.

1. Field Supplied Fan Sheave BK90 Required. Field Supplied Belt may be necessary.
2. 5 HP Oversize Motor with Field Supplied Motor Sheave 1VL44 Required.
   Field Supplied Belt may be necessary.
3. 5 HP Oversize Motor with Field Supplied Fan Sheave BK120 Required.
   Field Supplied Belt may be necessary.

Performance Specifications

Static Pressure Drops (Inches Water Column)

	Economizer Pressure Drop (Inches of W.C.)		
	100% Return Air	100% Outside Air	
CFM	Downflow/ Horizontal	Downflow	Horizontal
2700	0.07/ 0.03	0.13	0.13
3200	0.09/ 0.04	0.15	0.15
3400	0.11/ 0.04	0.16	0.16
4000	0.15/ 0.07	0.18	0.18
4800	0.22/ 0.11	0.23	0.23

Motor Sheave / Fan Speed

	Fan Speed (RPM)		
Motor Sheave Position	Standard Motor & Sheave	Standard Motor & High Static Drive	Oversized Motor & Sheave Kit DF/ HZ
6 Turns Open	958	*	1121/ 1032
5 Turns Open	1028		1180/ 1086
4 Turns Open	1081		1239/ 1141
3 Turns Open	1133		1297/ 1195
2 Turns Open	1186		1356/ 1250
1 Turn Open	1210		1415/ 1304
Closed	N/A		N/A

* Field supplied 1VP56 for Downflow units only.

Filter Pressure Drop

	Filter Pressure Drop (Inches of W.C.)				
	Standard Filters	Area (Sq. Ft.)		Velocity (Ft. per Min.)	
CFM	Downflow/ Horizontal	Downflow	Horizontal	Downflow	Horizontal
3200	0.05/ 0.08	10.0	12.5	320	256
3600	0.06/ 0.10			360	288
4000	0.06/ 0.12			400	320
4400	0.07/ 0.14			440	352
4800	0.08/ 0.16			480	384

Gas Heater Operating Data

Heating Input Rate – BTUH [3]		135,000	205,000
Minimum Supply Gas Pressure	Natural/ LP	3.5” w.c./ 8.0” w.c.	
Manifold Gas Pressure **(See Note 1)**		-0.2” w.c	
Combustion Blower Suction Pressure	(1st Stage)	-2.1 to -3.1” w.c.	-0.8 to -1.2” w.c.
(With Gas Valve Closed)	(2nd Stage	N/A	-2.1 to -3.1” w.c.
Minimum Flame Sensing Current **(See Note 2)**		5.0 Microamps D.C.	
Normal Sensing Current Range		8.0 to 16.0 Microamps D.C.	
Flue Gas Temperature Rise Above Ambient		400 to 500 F	350 to 475 F
Flue Gas Content - %CO2	Natural	8.3 to 9.5	8.0 to 9.0
	LP	9.5 to 10.5	
Minimum Supply Air Temperature Across Heat Exchanger		40 F	

Notes:

1. This Unit has a negative pressure gas valve. Never adjust to a positive pressure.
2. A voltage reading across pens (V+) & (V-) is equatable to the flame sensing current. One volt equals one micro amp.
3. For 50 Hertz applications, multiply rated BTUH by 83 percent.

Mechanical Data

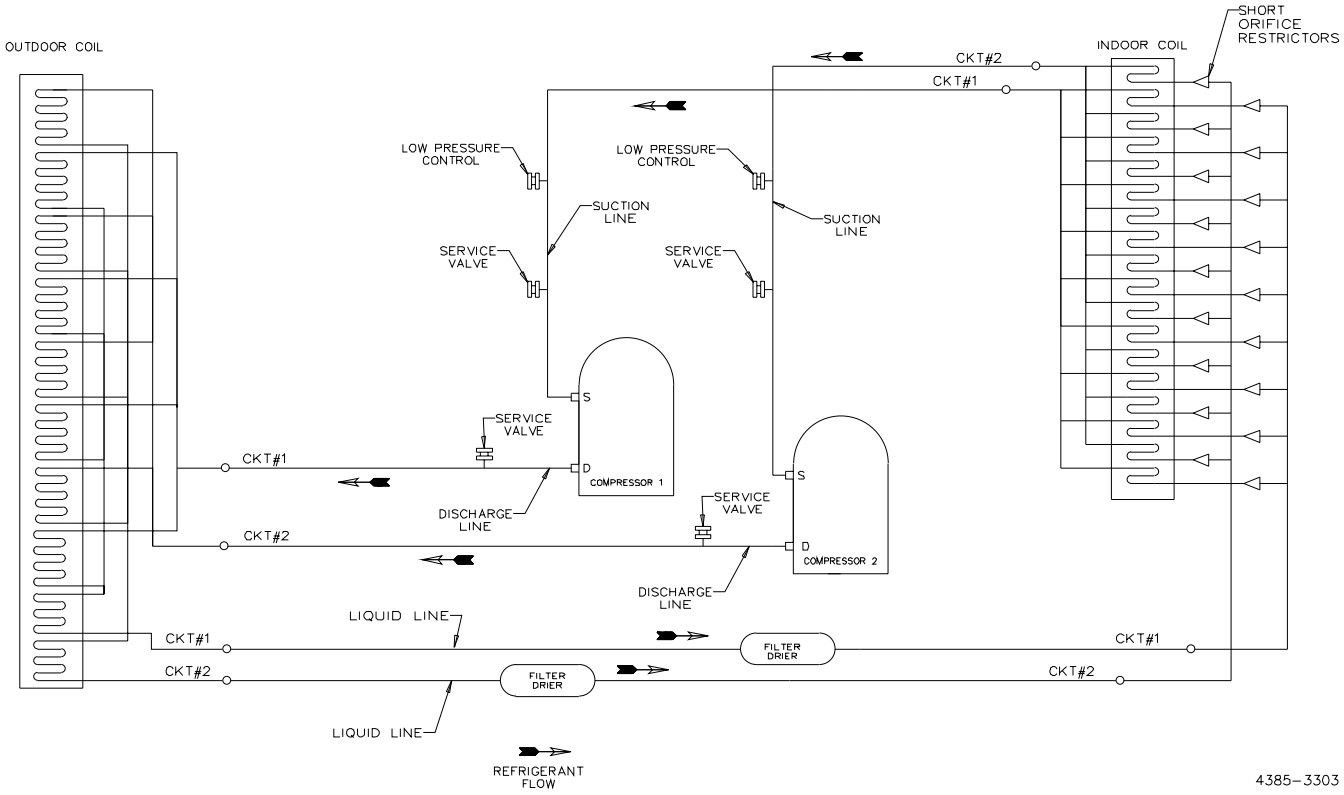
Refrigerant Circuit Diagram

⚠ CAUTION

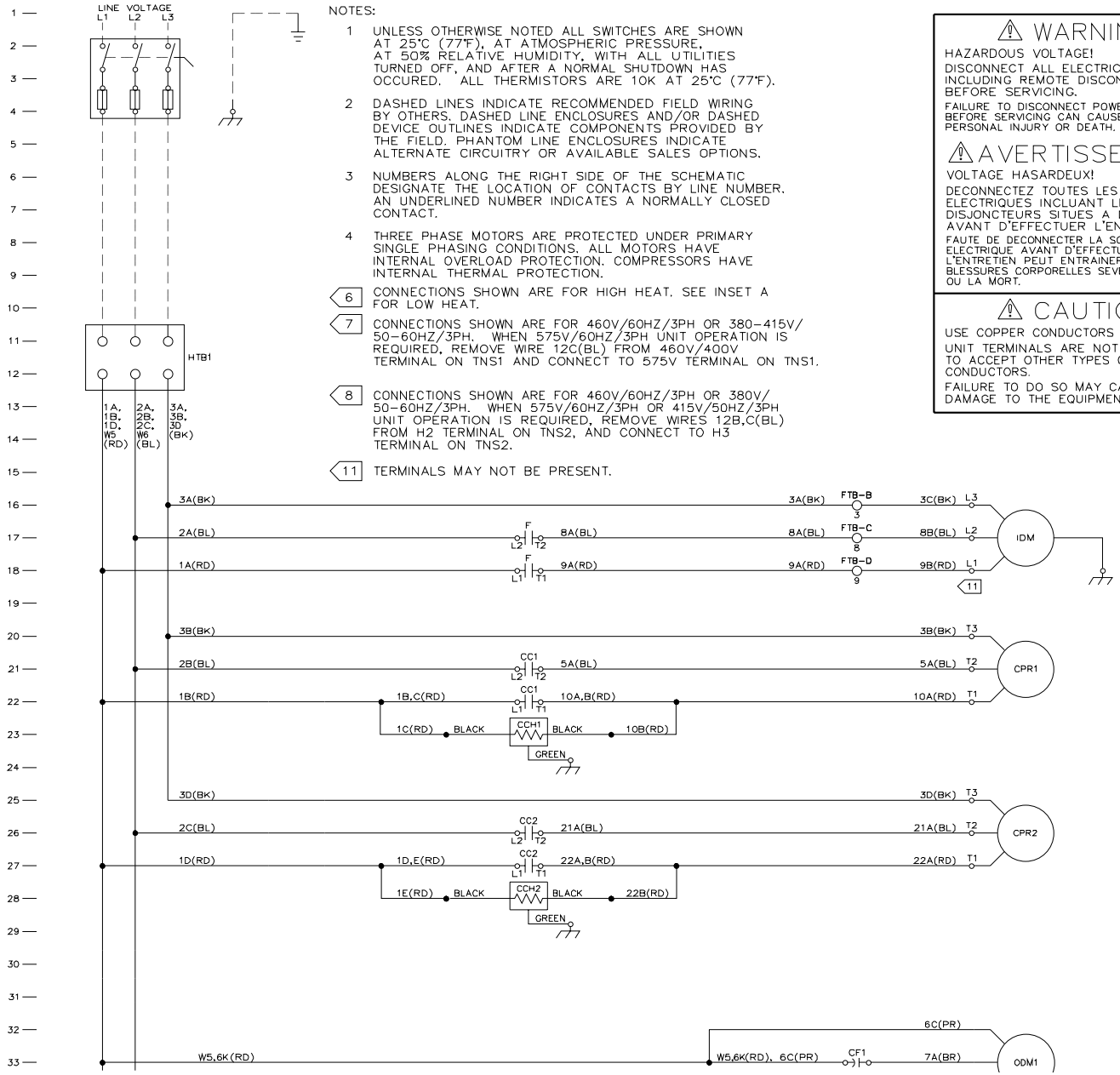
CONTAINS REFRIGERANT!

SYSTEM CONTAINS OIL AND REFRIGERANT UNDER HIGH PRESSURE. RECOVER REFRIGERANT TO RELIEVE PRESSURE BEFORE OPENING THE SYSTEM.

Failure to follow proper procedures can result in personal illness or injury or severe equipment damage.



Power Schematic



NOTES:

1. UNLESS OTHERWISE NOTED ALL SWITCHES ARE SHOWN AT 25°C (77°F), AT ATMOSPHERIC PRESSURE, AT 50% RELATIVE HUMIDITY, WITH ALL UTILITIES TURNED OFF, AND AFTER A NORMAL SHUTDOWN HAS OCCURED. ALL THERMISTORS ARE 10K AT 25°C (77°F).
2. DASHED LINES INDICATE RECOMMENDED FIELD WIRING BY OTHERS. DASHED LINE ENCLOSURES AND/OR DASHED DEVICE OUTLINES INDICATE COMPONENTS PROVIDED BY THE FIELD. PHANTOM LINE ENCLOSURES INDICATE ALTERNATE CIRCUITRY OR AVAILABLE SALES OPTIONS.
3. NUMBERS ALONG THE RIGHT SIDE OF THE SCHEMATIC DESIGNATE THE LOCATION OF CONTACTS BY LINE NUMBER. AN UNDERLINED NUMBER INDICATES A NORMALLY CLOSED CONTACT.
4. THREE PHASE MOTORS ARE PROTECTED UNDER PRIMARY SINGLE PHASING CONDITIONS. ALL MOTORS HAVE INTERNAL OVERLOAD PROTECTION. COMPRESSORS HAVE INTERNAL THERMAL PROTECTION.

<6> CONNECTIONS SHOWN ARE FOR HIGH HEAT. SEE INSET A FOR LOW HEAT.

<7> CONNECTIONS SHOWN ARE FOR 460V/60HZ/3PH OR 380-415V/50-60HZ/3PH. WHEN 575V/60HZ/3PH UNIT OPERATION IS REQUIRED, REMOVE WIRE 12C(BL) FROM 460V/400V TERMINAL ON TNS1 AND CONNECT TO 575V TERMINAL ON TNS1.

<8> CONNECTIONS SHOWN ARE FOR 460V/60HZ/3PH OR 380V/50-60HZ/3PH. WHEN 575V/60HZ/3PH OR 415V/50HZ/3PH UNIT OPERATION IS REQUIRED, REMOVE WIRES 12B,C(BL) FROM H2 TERMINAL ON TNS2, AND CONNECT TO H3 TERMINAL ON TNS2.

<11> TERMINALS MAY NOT BE PRESENT.

⚠ WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRIC POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

FAILURE TO DISCONNECT POWER BEFORE SERVICING CAN CAUSE SEVERE PERSONAL INJURY OR DEATH.

⚠ AVERTISSEMENT

VOLTAGE HASARDEUX!

DECONNECTEZ TOUTES LES SOURCES ELECTRIQUES INCLUANT LES DISJONCTEURS SITUES A DISTANCE AVANT D'EFFECTUER L'ENTRETIEN.

FAUTE DE DECONNECTER LA SOURCE ELECTRIQUE AVANT D'EFFECTUER L'ENTRETIEN PEUT ENTRAINER DES BLESSURES CORPORELLES SEVERES OU LA MORT.

⚠ CAUTION

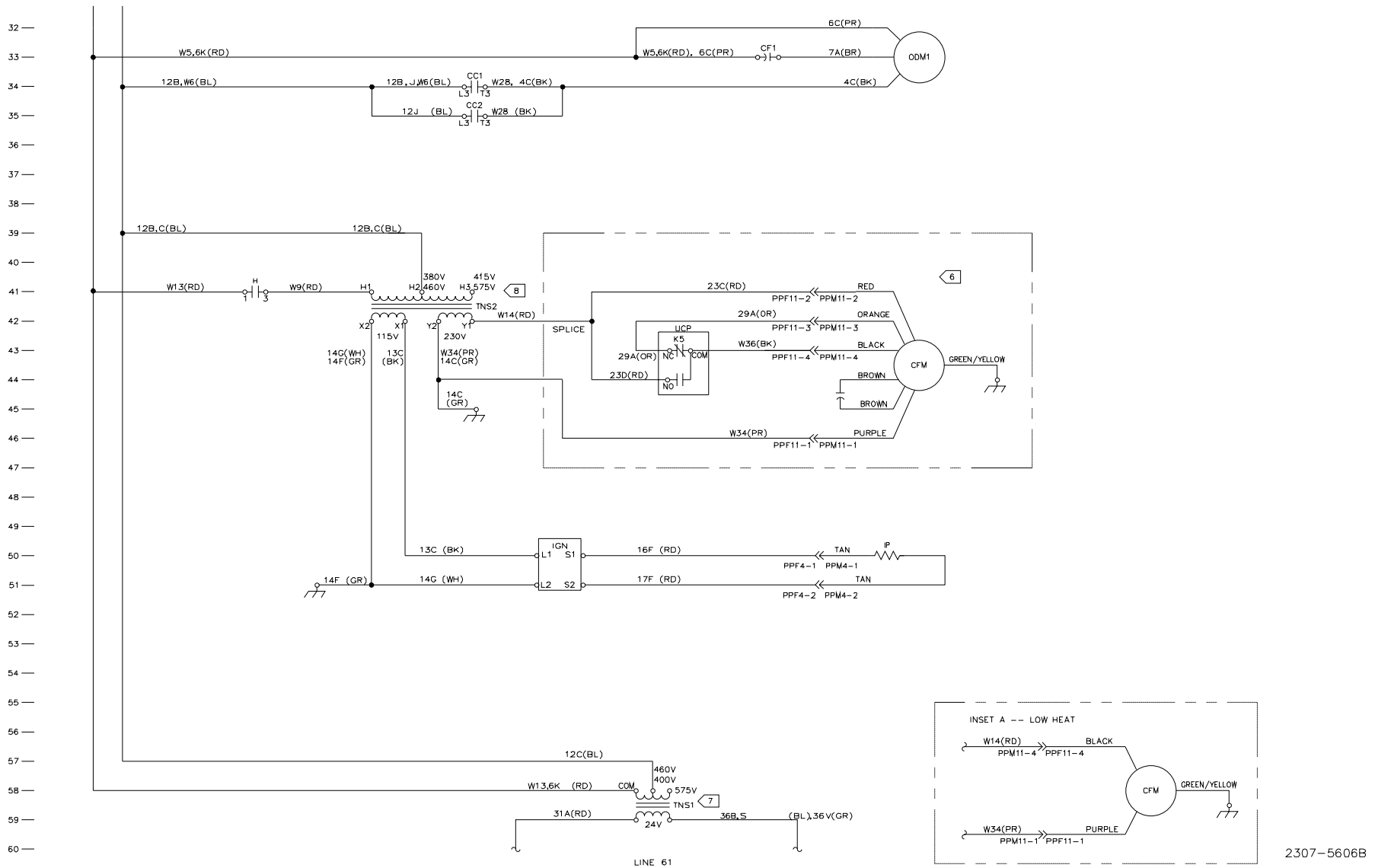
USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

FAILURE TO DO SO MAY CAUSE DAMAGE TO THE EQUIPMENT.

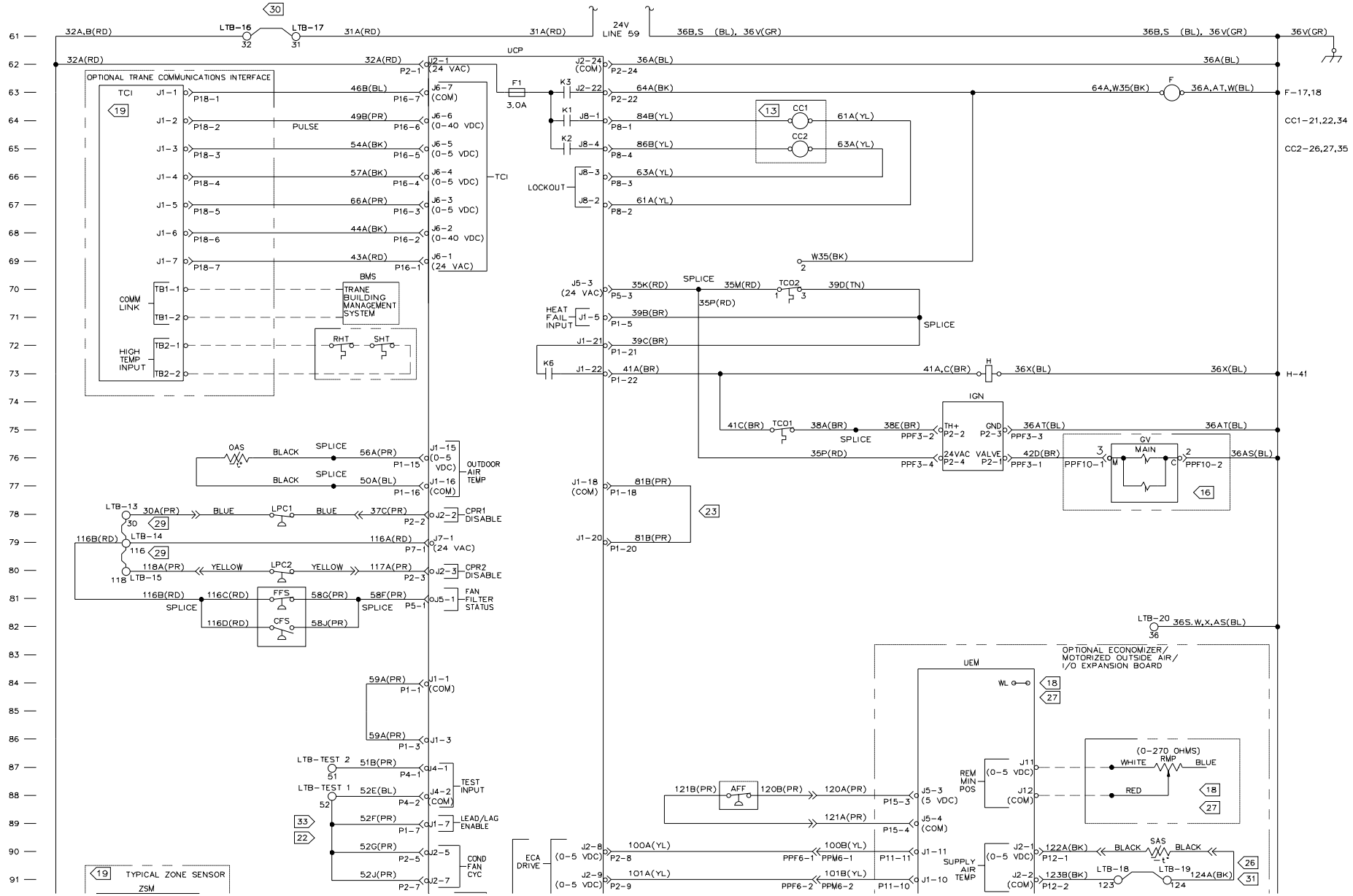
LEGEND		
DEVICE DESIGNATION	DESCRIPTION	LINE NUMBER
AFF	ACTIVE FAN FAILURE	88
BMS	BUILDING MANAGEMENT SYSTEM	70
CC1,CC2	COMPRESSOR CONTACTOR	64,65
CCH1,CCH2	CRANKCASE HEATER	23,28
CF1	OUTDOOR MOTOR CAPACITOR	33
CFM	COMBUSTION FAN MOTOR	43,58
CFS	CLOGGED FILTER SWITCH	82
CPR1,CPR2	COMPRESSOR	21,26
ECA	ECONOMIZER ACTUATOR	99
FFS	FAN FAILURE SWITCH	81
F	INDOOR FAN CONTACTOR	63
F1	UCP FUSE	63
FTB	FAN TERMINAL BLOCK	16,17,18
GV	GAS VALVE	76
HPC1,HPC2	HIGH PRESSURE CONTROL	107,109
H	HEAT RELAY	73
HTB1	HI VOLTAGE TERMINAL BLOCK	11
IDM	INDOOR FAN MOTOR	17
IGN	IGNITION CONTROL MODULE	74,50
IP	IGNITION PROBE	50
LTB	LOW VOLTAGE TERM BLOCK	
LPC1,LPC2	LOW PRESSURE CONTROL	78,80
OAS	OUTDOOR AIR SENSOR	76
OHS	OUTDOOR HUMIDITY SENSOR	96
ODM1	OUTDOOR FAN MOTOR	33
RAS	RETURN AIR SENSOR	92
RHS	RETURN HUMIDITY SENSOR	94
RHT	RETURN HIGH TEMPERATURE	72
RMP	REMOTE MINIMUM POSITION	87
SAS	SUPPLY AIR SENSOR	91
SHT	SUPPLY HIGH TEMPERATURE	72
TCI	TRANE COMMUNICATION INTERFACE	63
TCO1	HIGH LIMIT CUTOUT	75
TCO2	FAN FAILURE LIMIT	70
TNS1	CONTROL POWER TRANSFORMER	58
TNS2	IGNITION TRANSFORMER	41
UEM	UNITARY ECONOMIZER MODULE	87
UCP	UNITARY CONTROL PROCESSOR	62
XFR	EXHAUST FAN RELAY	99
ZSM	ZONE SENSOR MODULE	92
K5	UCP HEAT RELAY	43
PPF6,PPM6	ECONOMIZER PLUG	90-102
PPF3	IGNITION CONTROL PLUG	50,75
PPF4,PPM4	IGNITER PLUG	50
PPF11,PPM11	COMBUSTION FAN MOTOR PLUG	41-46

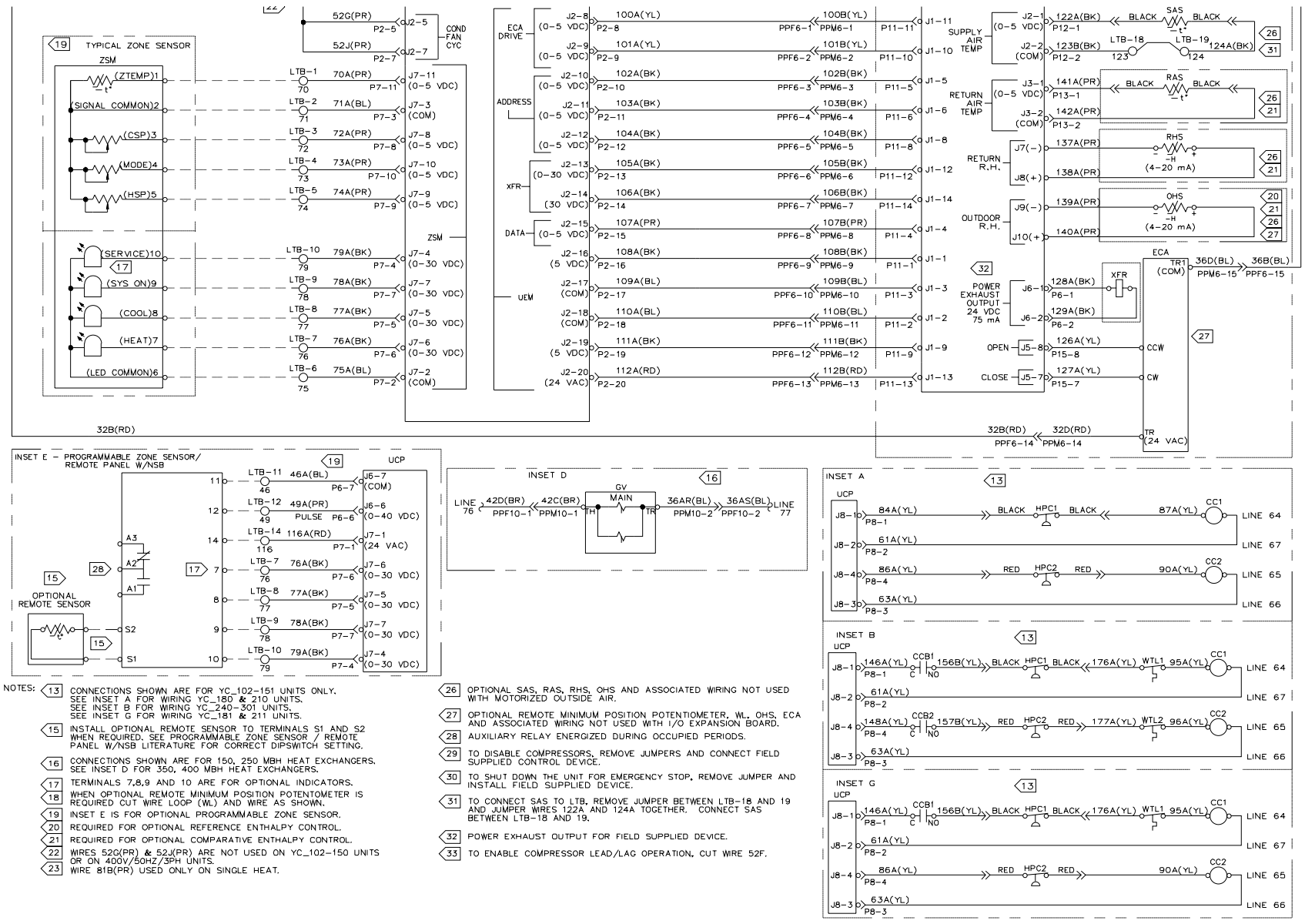
WIRE COLOR DESIGNATION			
ABBR	COLOR	ABBR	COLOR
BK	BLACK	PR	PURPLE
BL	BLUE	RD	RED
BR	BROWN	WH	WHITE
GR	GREEN	YL	YELLOW
OR	ORANGE	TN	TAN


6C(PR)
W5,6K(RD)
W5,6K(RD), 6C(PR)
CF1
7A(BR)
ODM1
12B,W6(BL)
12B,J,W6(BL)
CC1
W28, 4C(BK)
4C(BK)
12J (BL)
CC2
W28 (BK)
L3
T3
12B,C(BL)
12B,C(BL)
W13(RD)
H
W9(RD)
H1
H2
H3
380V
415V
460V
575V
8
TNS2
W14(RD)
X2
X1
Y2
Y1
115V
230V
14G(WH)
14F(GR)
13C
(BK)
W34(PR)
14C(GR)
14C
(GR)
SPLICE
6
23C(RD)
RED
PPF11-2
PPM11-2
29A(OR)
ORANGE
PPF11-3
PPM11-3
UCP
K5
NC
COM
NO
W36(BK)
BLACK
PPF11-4
PPM11-4
CFM
GREEN/YELLOW
BROWN
BROWN
W34(PR)
PURPLE
PPF11-1
PPM11-1
IGN
L1
S1
L2
S2
13C (BK)
16F (RD)
TAN
IP
PPF4-1
PPM4-1
14F (GR)
14G (WH)
17F (RD)
TAN
PPF4-2
PPM4-2
12C(BL)
460V
400V
575V
W13,6K (RD)
COM
7
TNS1
31A(RD)
24V
36B,S
(BL),36V(GR)
LINE 61
INSET A -- LOW HEAT
W14(RD)
BLACK
PPM11-4
PPF11-4
CFM
GREEN/YELLOW
W34(PR)
PURPLE
PPM11-1
PPF11-1

2307-5606B

Control Schematic





NOTES:

13 CONNECTIONS SHOWN ARE FOR YC_102-151 UNITS ONLY. SEE INSET A FOR WIRING YC_180 & 210 UNITS. SEE INSET B FOR WIRING YC_240-301 UNITS. SEE INSET G FOR WIRING YC_181 & 211 UNITS.

15 INSTALL OPTIONAL REMOTE SENSOR TO TERMINALS S1 AND S2 WHEN REQUIRED. SEE PROGRAMMABLE ZONE SENSOR / REMOTE PANEL W/NSB LITERATURE FOR CORRECT DIPSWITCH SETTING.

16 CONNECTIONS SHOWN ARE FOR 150, 250 MBH HEAT EXCHANGERS. SEE INSET D FOR 350, 400 MBH HEAT EXCHANGERS.

17 TERMINALS 7,8,9 AND 10 ARE FOR OPTIONAL INDICATORS.

18 WHEN OPTIONAL REMOTE MINIMUM POSITION POTENTOMETER IS REQUIRED CUT WIRE LOOP (WL) AND WIRE AS SHOWN.

19 INSET E IS FOR OPTIONAL PROGRAMMABLE ZONE SENSOR.

20 REQUIRED FOR OPTIONAL REFERENCE ENTHALPY CONTROL.

21 REQUIRED FOR OPTIONAL COMPARATIVE ENTHALPY CONTROL.

22 WIRES 52G(PR) & 52J(PR) ARE NOT USED ON YC_102-150 UNITS OR ON 400V/50HZ/3PH UNITS.

23 WIRE 81B(PR) USED ONLY ON SINGLE HEAT.

26 OPTIONAL SAS, RAS, RHS, OHS AND ASSOCIATED WIRING NOT USED WITH MOTORIZED OUTSIDE AIR.

27 OPTIONAL REMOTE MINIMUM POSITION POTENTIOMETER, WL, OHS, ECA AND ASSOCIATED WIRING NOT USED WITH I/O EXPANSION BOARD.

28 AUXILIARY RELAY ENERGIZED DURING OCCUPIED PERIODS.

29 TO DISABLE COMPRESSORS, REMOVE JUMPERS AND CONNECT FIELD SUPPLIED CONTROL DEVICE.

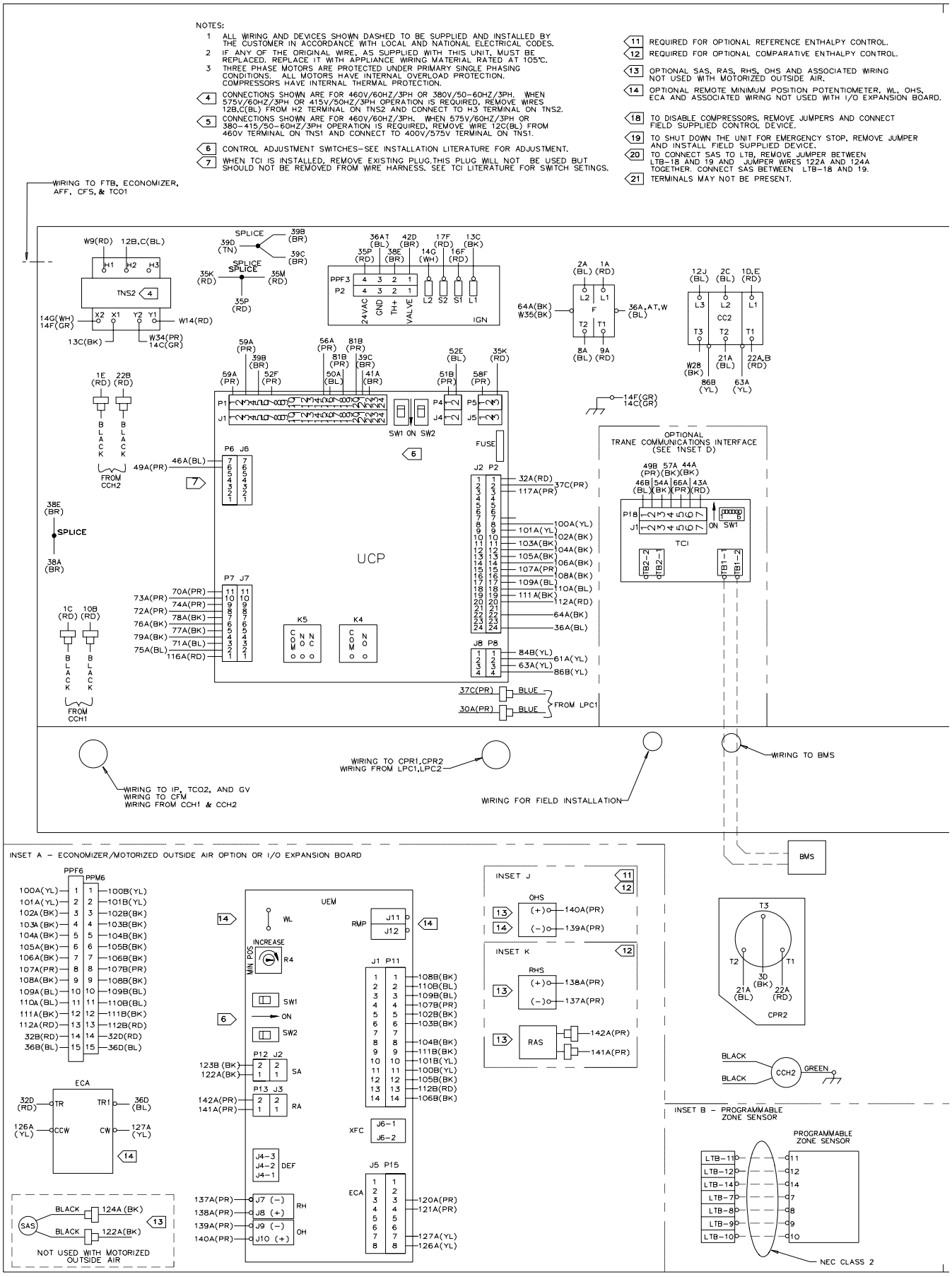
30 TO SHUT DOWN THE UNIT FOR EMERGENCY STOP, REMOVE JUMPER AND INSTALL FIELD SUPPLIED DEVICE.

31 TO CONNECT SAS TO LTB, REMOVE JUMPER BETWEEN LTB-18 AND 19 AND JUMPER WIRES 122A AND 124A TOGETHER. CONNECT SAS BETWEEN LTB-18 AND 19.

32 POWER EXHAUST OUTPUT FOR FIELD SUPPLIED DEVICE.

33 TO ENABLE COMPRESSOR LEAD/LAG OPERATION, CUT WIRE 52F.

Connections Diagram (Low Heat)



IMPORTANT!
DO NOT ENERGIZE
UNIT UNTIL CHECK-OUT
AND START-UP PROCEDURE
HAS BEEN COMPLETED

FUSE REPLACEMENT TABLE			
UCP	ALL	TYPE MDL	3.0 AMP

WIRE COLOR DESIGNATION			
ABBR	COLOR	ABBR	COLOR
BK	BLACK	PR	PURPLE
BL	BLUE	RD	RED
BR	BROWN	WH	WHITE
GR	GREEN	YL	YELLOW
OR	ORANGE	TN	TAN

⚠ WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRIC POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

FAILURE TO DISCONNECT POWER BEFORE SERVICING CAN CAUSE SEVERE PERSONAL INJURY OR DEATH.

⚠ AVERTISSEMENT

VOLTAGE HASARDEUX!

DECONNECTEZ TOUTES LES SOURCES ELECTRIQUES INCLUANT LES DISJONCTEURS SITUES A DISTANCE AVANT D'EFFECTUER L'ENTRETIEN.

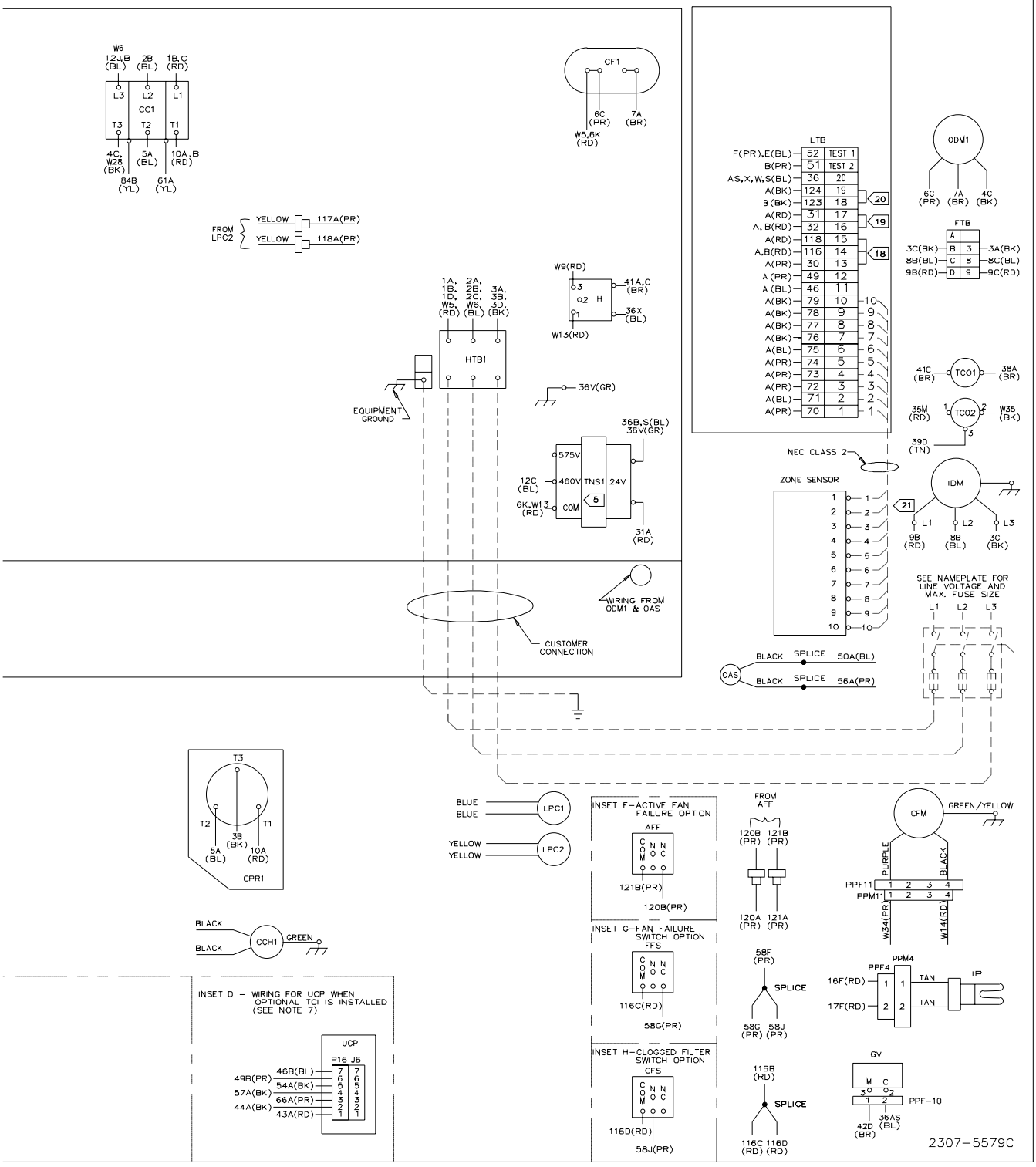
FAUTE DE DECONNECTER LA SOURCE ELECTRIQUE AVANT D'EFFECTUER L'ENTRETIEN PEUT ENTRAINER DES BLESSURES CORPORELLES SEVERES OU LA MORT.

⚠ CAUTION

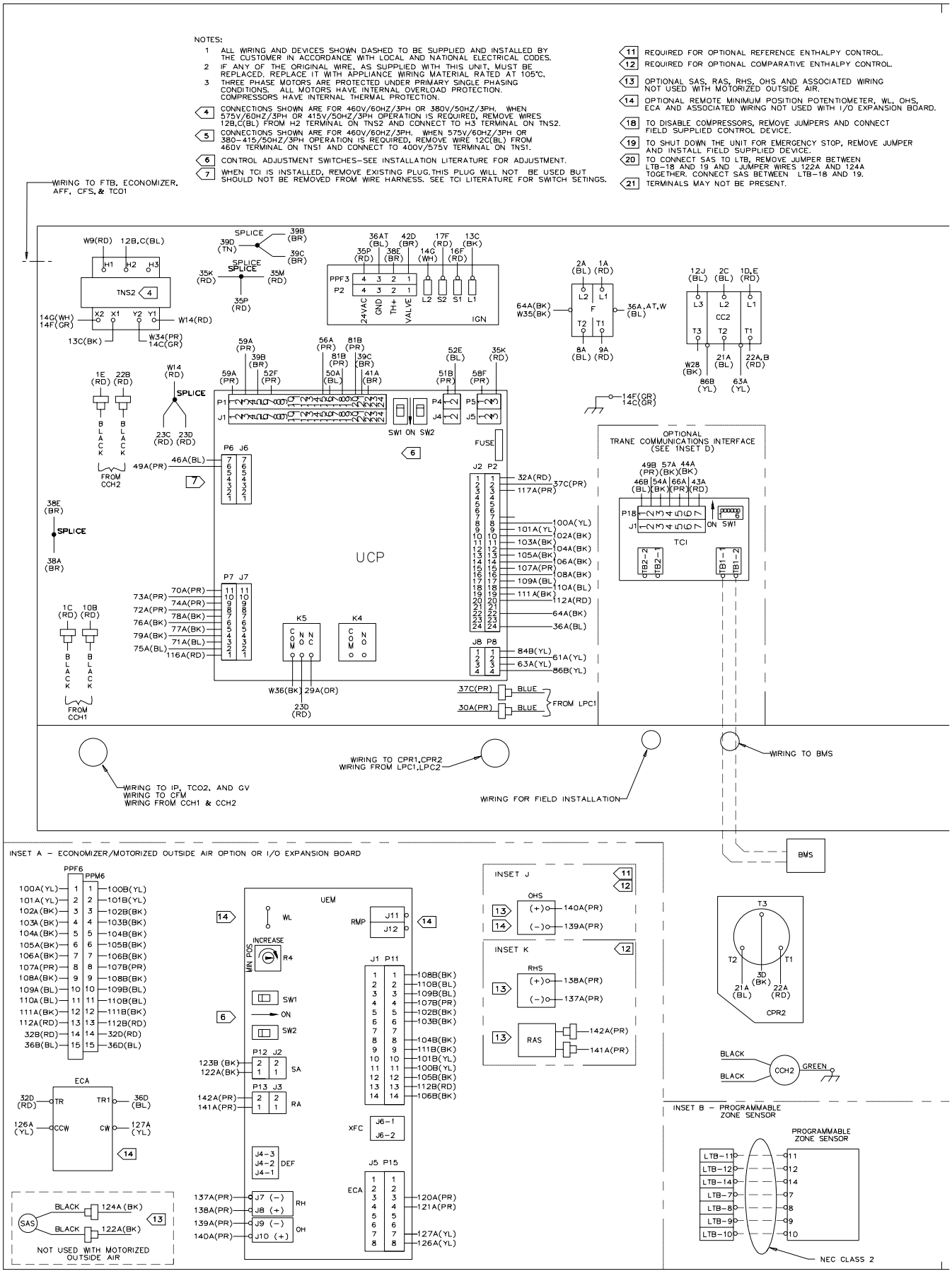
USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

FAILURE TO DO SO MAY CAUSE DAMAGE TO THE EQUIPMENT.



Connections Diagram (Medium Heat)



IMPORTANT!
DO NOT ENERGIZE UNIT UNTIL CHECK-OUT AND START-UP PROCEDURE HAS BEEN COMPLETED

FUSE REPLACEMENT TABLE			
UCP	ALL	TYPE MDL	3.0 AMP

WIRE COLOR DESIGNATION			
ABBR	COLOR	ABBR	COLOR
BK	BLACK	PR	PURPLE
BL	BLUE	RD	RED
BR	BROWN	WH	WHITE
GR	GREEN	YL	YELLOW
OR	ORANGE	TN	TAN

⚠ WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRIC POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

FAILURE TO DISCONNECT POWER BEFORE SERVICING CAN CAUSE SEVERE PERSONAL INJURY OR DEATH.

⚠ AVERTISSEMENT

VOLTAGE HASARDEUX!

DECONNECTEZ TOUTES LES SOURCES ELECTRIQUES INCLUANT LES DISJONCTEURS SITUES A DISTANCE AVANT D'EFFECTUER L'ENTRETIEN.

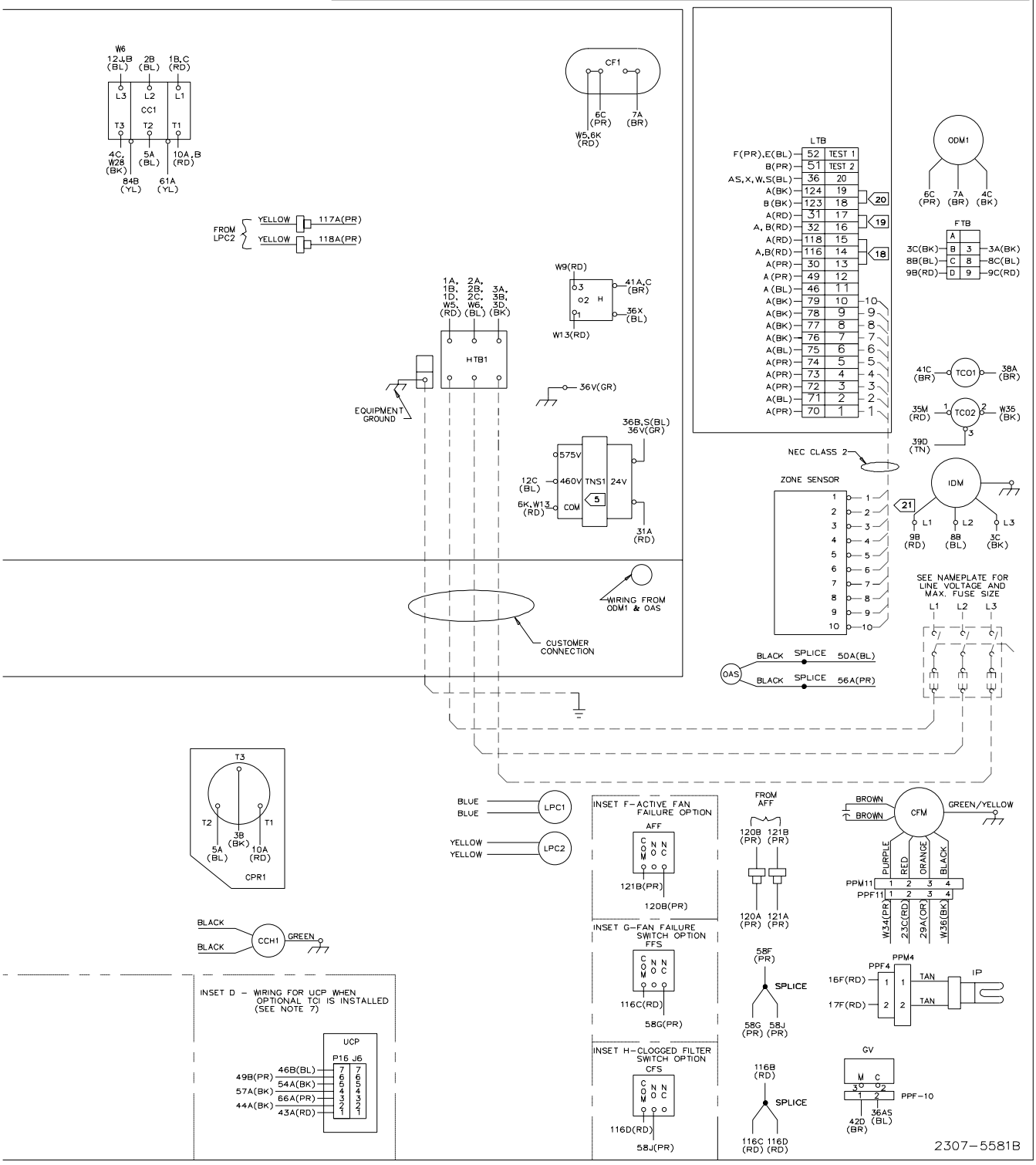
FAUTE DE DECONNECTER LA SOURCE ELECTRIQUE AVANT D'EFFECTUER L'ENTRETIEN PEUT ENTRAINER DES BLESSURES CORPORELLES SEVERES OU LA MORT.

⚠ CAUTION

USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

FAILURE TO DO SO MAY CAUSE DAMAGE TO THE EQUIPMENT.



Sequence of Operation

These units are equipped with a microelectronic control feature which provides operating functions that are significantly different than conventional units. It is referred to as the “Unit Control Processor” **(UCP)**.

The UCP provides compressor anti short cycle timing functions through minimum “Off” and “On” timing to increase reliability, performance and to maximize unit efficiency.

Upon power initialization, the UCP performs self-diagnostic checks to insure that all internal controls are functioning. It checks the configuration parameters against the components connected to the system. When units are equipped with an optional economizer, the UCP drives the economizer dampers toward the open position for 15 to 20 seconds then closes. The economizer is closed for approximately 90 seconds to ensure proper damper calibration.

The LED located on the UCP module is turned “On” within one second after power-up if all internal operations are okay.

Lead/Lag Operation
The UCP can be configured for Lead/Lag operation. To enable lead/lag, wire 52F **(PR)** connected to the J1-7 terminal on the UCP must be cut. Refer to the unit wiring diagram for the terminal location. Once configured, the UCP will alternate the starting of the circuit one and circuit two compressor after each cooling request has been satisfied.

Note: Upon each power initialization, the UCP will start the number one circuit compressor. When a Coventional Thermostat Interface (CTI) is installed, the UCP will not allow lead/lag operation during the TEST mode.

Cooling without an Economizer (Lead/Lag Disabled)
When the system switch is set to the “Cool” position, and the zone temperature rises above the cooling setpoint controlband, the UCP energizes the first stage relay **(K1)** coil located on the UCP. When the K1 relay contacts close, the compressor contactor coil **(CC1)** is energized provided the low pressure control **(LPC1)** is closed. When the CC1 contacts close, compressor **(CPR1)** and the outdoor fan motor **(ODM)** start to maintain the zone temperature to within ±2 F of the sensor setpoint at the sensed location.

In addition to starting the compressor **(CPR1)** and outdoor fan motor **(ODM)** when the **(CC1)** contacts close, the compressor crankcase heater **(CCH1)** is de-energized through redirection of curent flow through the CC1 contacts.

If additional cooling is required, the UCP will energize the second stage relay **(K2)** coil to energize the compressor contactor **(CC2)** coil, provided the low pressure control **(LPC2)** is closed, to start the second compressor **(CPR2)** and de-energize the compressor crankcase heater **(CCH2)** through redirection of curent flow through the CC2 contacts.

As the cooling requirement is satisfied, the UCP cycles the compressor(s) and outdoor fan motor “Off”. Current is allowed to flow through each crankcase heater **(CCH1 & CCH2)** as **CC1** and **CC2** contacts open.

Evaporator Fan Operation
When the fan selection switch is set to the “Auto” position, the UCP energizes the **(K3)** relay coil approximately 1 second after energizing the compressor contactor coil **(CC1)** in the cooling mode. Closing the **K3** contacts on the UCP energizes the indoor fan relay **(F)** coil to start the indoor fan motor **(IDM)**.

The UCP de-energizes the fan relay **(F)** approximately 60 seconds after the cooling requirement has be satisfied to enhance unit efficiency.

When the fan selection switch is set to the “On” position, the UCP keeps the indoor fan relay coil **(F)** energized for continuous fan motor operation.

When the unit is equipped with the optional fan failure switch, wired between terminals J5-1 and J7-1 on the **UCP**, the **UCP** produces an analog output if the fan failure switch **(FFS)** does not open within two minutes after a request for fan operation. When the system is connected to a remote panel, the “SERVICE” LED will be turned on when this failure occurs.

Low Ambient Operation
During low ambient operation, outside air temperature below 55 F, the UCP will cycle the compressor and outdoor fan motor “Off” for approximately 3 minutes after every 10 minutes of accumulated compressor run time. The indoor fan motor **(IDM)** will continue to operate during this evaporator defrost cycle **(EDC)** and the compressor and outdoor fan will return to normal operation once the defrost cycle has terminated and the compressor “Off” time delay has been satisfied.

Cooling with an Economizer
The economizer is utilized to control the zone temperature providing the outside air conditions are suitable. Outside air is drawn into the unit through modulating dampers. When cooling is required and economizing is possible, the UCP

sends the cooling request to the unit economizer module **(UEM)** to open the economizer dampers by energizing the economizer actuator **(ECA)**. The UCP tries to cool the zone utilizing the economizer to slightly below the zone temperature setpoint. If the supply air sensor **(SAS)** senses that the supply air temperature is below 50 F, the dampers modulated toward the closed position. If the zone temperature continues to rise above the zone temperature setpoint controlband and the economizer dampers are less than 80% of full open, the UCP energizes the first stage of mechanical cooling. If the zone temperature continues to rise above the zone temperature setpoint controlband, the UCP energizes the second stage of mechanical cooling.

Note: Second stage mechanical cooling is locked out when units are equipped with the conventional thermostat interface (CTI) during economizer operation.

During simultaneous economizing and mechanical cooling (compressor operation), the UEM continues to modulate the economizer dampers open/closed to keep the supply air temperature in the 50 F to 55 F range with the compressor(s) operating.

If economizing is not possible, the UEM drives the dampers to the minimum position setpoint when the indoor fan relay **(F)** is energized and allows mechanical cooling operation.

When the unit is equipped with the optional fan failure switch, wired between terminals J5-3 and J5-4 on the **UEM**, the **UCP** will stop all cooling functions and produce an analog output if the active fan failure switch **(AFF)** does not open within 40 seconds after a request for fan operation. When the system is connected to a remote panel, the "SERVICE" LED will flash when this failure occurs.

Economizer Set-Up
The required amount of ventilation air is set by adjusting the minimum position potentiometer located on the unit economizer module **(UEM)**.

Two of the three methods for determining the suitability of the outside air can be selected utilizing the DIP switches on the UEM, as described below:

1. Ambient Temperature - controlling the economizing cycle by sensing the outside air dry bulb temperature. The Table below lists the selectable dry bulb values by DIP switch setting.
2. Reference Enthalpy - controlling the economizer cycle by sensing the outdoor air humidity. The Table below lists the selectable enthalpy values by DIP switch setting. If the outside air enthalpy value is less than the selected value, the economizer is allowed to operate.
3. Comparative Enthalpy - By utilizing a humidity sensor and a temperature sensor in both the return air stream and the outdoor air stream, the unit control processor **(UCP)** will be able to establish which conditions are best suited for maintaining the zone temperature, i.e. indoor conditions or outdoor conditions. The DIP switches located on the UEM are non-functional when both the temperature and humidity sensors are installed.

Selection	Dry Bulb	Enthalpy	SW1	SW2
A	60 F	19 BTU/LB	OFF	OFF
B	55 F	22 BTU/LB	OFF	ON
C	65 F	25 BTU/LB	ON	OFF
D	-----	28 BTU/LB	ON	ON

Heating Operation (Low Heat)
When the system switch is set to the "Heat" position and the zone temperature falls below the heating setpoint controlband, a heat cycle is initiated when the **K6** relay coil on the UCP is energized. The normally open **K6** relay contacts on the UCP closes energizing the heat relay **(H)** coil and powers the 24 VAC ignition control module **(IGN)** provided **TCO1** is closed. When the normally open heat relay **(H)** contacts (1 & 3) closes, power is supplied to the ignition transformer **(TNS2**) powering the 115 VAC section of the ignition control module **(IGN)** and starting the combustion fan motor **(CFM).**

Heating Operation (High Heat)
When the system switch is set to the "Heat" position and the zone temperature falls below the heating setpoint controlband, a heat cycle is initiated when the **K6** and **K5** relay coils on the UCP are energized. The normally open **K6** relay contacts on the UCP closes energizing the heat relay **(H)** coil and powers the 24 VAC ignition control module **(IGN)** provided **TCO1** is closed. When the normally open heat relay **(H)** contacts close, power is supplied to the ignition transformer **(TNS2)** powering the 115 VAC section of the ignition control module **(IGN)**. The normally open **K5** relay contacts on the UCP closes to start the combustion fan motor **(CFM)** in high speed. After approximately 60 seconds if no additional heating is required, the UCP de-energizes the **K5** relay coil and the combustion fan motor switches to low speed through the normally closed **K5** relay contacts.

Sequence of Operation

Ignition Module
With 115 VAC power supplied to the ignition module **(IGN)**, the hot surface ignition probe **(IP)** is preheated for approximately 30 to 45 seconds. The gas valve **(GV)** is energized for approximately 7 seconds for trial for ignition, to ignite the burner. Once the burner is ignited, the hot surface ignition probe **(IP)** is de-energized by the ignition module **(IGN)** and functions as the flame sensing device.

If the burner fails to ignite, the ignition module will attempt two retries. At the start of each ignition retry, the green LED will flash and the red LED will flash for five seconds before locking out. An ignition lockout can be reset by;

1. Opening and closing the main power disconnect switch,

2. By switching the “Mode” switch on the zone sensor to “OFF” and then to the desired position.

3. Allowing the ignition control module to reset automatically after one hour. Refer to the “Ignition Control Diagnostics” section for the LED diagnostic definitions.

When the fan selection switch is set to the “Auto” position, the UCP energizes the indoor fan relay **(F)** coil approximately 30 second after initiating the heating cycle to start the indoor fan motor **(IDM)**.

The automatic reset high limit **(TCO1)**, located in the bottom right corner of the burner compartment, protects against abnormally high leaving air temperatures. The automatic reset fan fail limit **(TCO2)**, located in the upper middle section of the indoor fan board, protects against abnormally high heat buildup which could occur because of extended cycling of the high limit **(TCO1)** or if the indoor fan motor **(IDM)** fails to operate. Should **TCO2** open, the UCP will de-energize the **K6** relay and energize the indoor fan relay **(F)** in an attempt to start the fan motor. The UCP signals that a heat failure has occurred by flashing the “Heat” LED on the zone sensor.

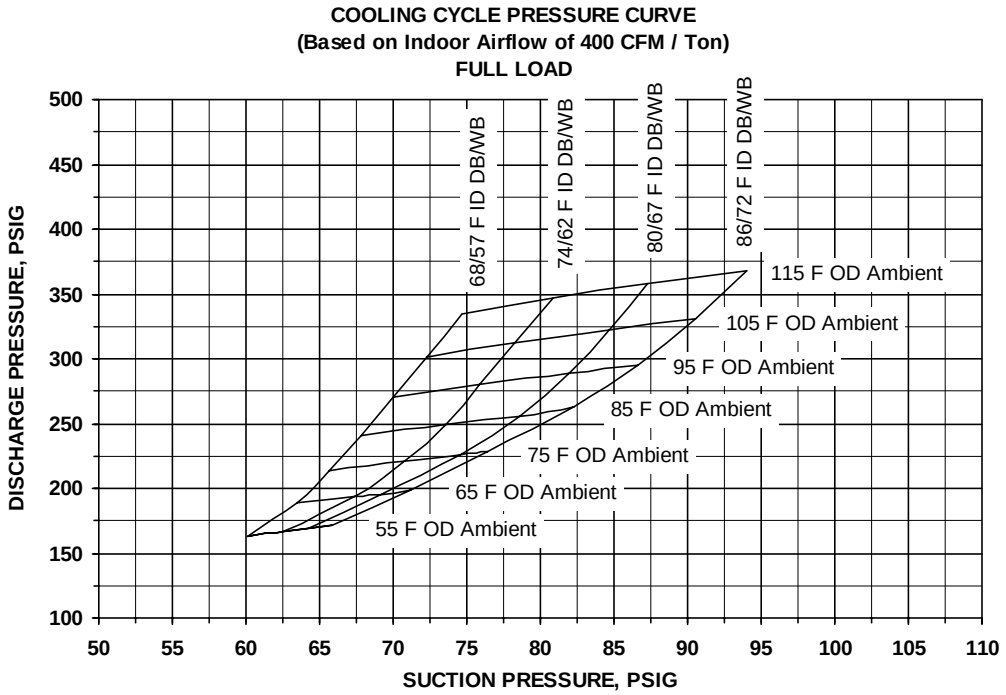
There are two LED’s located in the Ignition Control Module. The Table below lists the diagnostics and the status of the LED’s during the various operating states.

Ignition Control Module Diagnostics

Diagnostics	LED Status	
	Green LED	**Red LED**
1. Powered with no call for heat	Off	Off
2. Call for heat - no fault detected	Flashing (1)	Off
3. No Flame signal on try for ignition or flame signal established and lost prior to a lockout condition.	Off	Flashing (2)
4. Gas Valve miswired or flame signal present at a call for heat	Continuous	Flashing (1)
5. Internal Fault - anytime.	Off	Continuous

Notes:
1. Flash at a 50% duty cycle.
2. At the start of each retry for ignition the red LED will flash for five seconds along with the green LED

Refrigeration Data

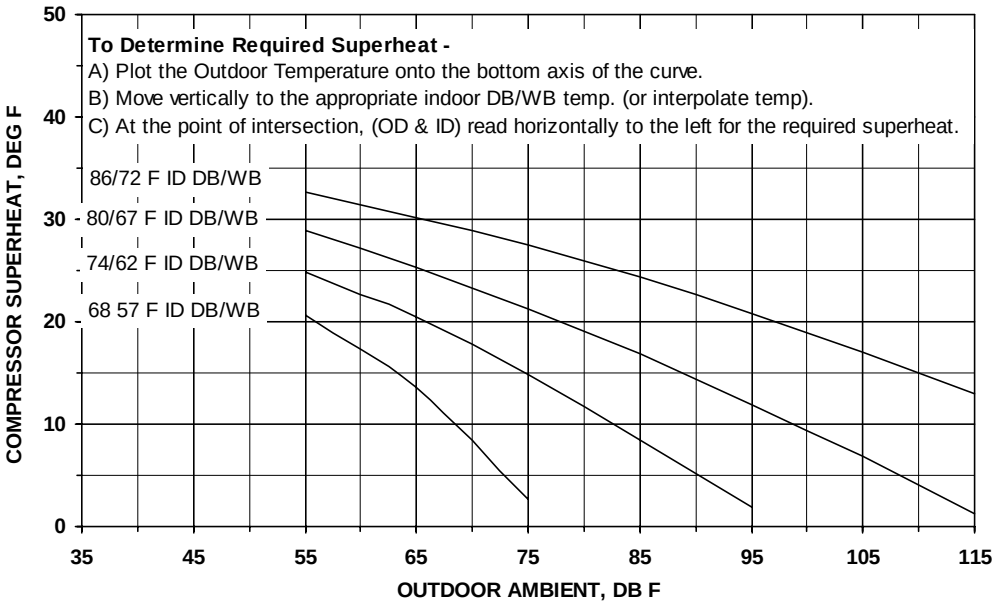


To Check Operating Pressures

1. Start the unit and allow the pressures to stabilize.
2. Measure the indoor DB/WB temperature entering the indoor coil.
3. Measure the outdoor air dry bulb temperature
4. Take discharge and suction pressure readings.
5. Plot the outdoor dry bulb and the indoor DB/WB temperature onto the chart.
6. At the point of intersection, read down for the suction pressure and to the left for the discharge pressure.

Refrigeration Data

SUPERHEAT CHARGING CHART
(Based on Indoor Airflow of 400 CFM / Ton)
FULL LOAD



1) REFRIGERANT CHARGE - ADD if the superheat is more than 5 F above curve value.
- REDUCE if the superheat is more than 5 F below curve value.
- OK if the superheat is within 5 F of curve value.

2) Do not add refrigerant if the superheat is less than 5 F.
3) Curves are based on 400 CFM/Ton Indoor Airflow @ 50% R.H.
4) System must be running at stablized conditions before measuring superheat.

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